

1. Explain any two of the following:

2x4

- Implementation of conditional execution of instructions.
- A scheme for synchronization of the processor with memory during read/ write operations.
- A scheme for reducing the length of a microinstruction.

2. Draw the schematic diagrams with appropriate labelling for any two of the following and explain their working:

2x4

- a Static RAM bit cell
- a Dynamic RAM bit cell
- a 4x4 EPROM module

3. a. What are the two forms of locality of reference observed in program execution and explain how cache memory utilizes these.

3

b. Explain the buffered write mechanism used in cache memory.

2

4. a. What are the advantages and shortcomings of paged virtual memory?

2

b. How does TLAB improve performance of paged virtual memory?

2

c. Explain with an appropriate schematic diagram the address translation mechanism for a segmentation based MMU.

3

5. Answer for 12 marks from the following:

a. A 512 Mbit DRAM chip is organized with 2 bits per location and equal number bits for row and column addresses.

- If the cells are required to be refreshed every 40ms and each refresh requires 10ns, what is the percentage of time (approx.) available for read/ write operations?

1½

ii. What is the size of the refresh buffer?

½

b. A computing hardware is to be designed with a processor generating 32bit address for memory. It requires 3GB of RAM, 256MB of EPROM and 4MB of EEPROM and these are mapped in the physical address space as follows.

EEPROM: 0 to (4M-1)
 EPROM: 256M to (512M-1)
 RAM: 1G to (4G-1)

Write the Boolean expressions for the signals- EEPROM Select, EPROM Select, RAM Select.

2

- The cache memory in a computer uses 8-way set-associative mapping. The cache size is 512 KB with cache lines of 16 bytes. The memory address length is 32 bits. Determine the total size (in bits) of the comparator required in the cache.
- A computer system has L1 and L2 cache with access times of 5ns and 10ns respectively. The main memory access time for a cache line is 100ns and the average access time of a page from online storage is 2ms. During execution of a computer program experiences hit ratios of 0.95 in L1, 0.98 in L2 and 0.998 in the main memory. Determine the average memory access time during the execution.
- A computer system uses a processor that generates 40-bit address. The system has 2 GB of main memory. The MMU uses pages of 512 bytes. What will be the total size of the page table?
- A DRAM memory chip is organized as (32k x 32k x 4) bit cells. Considering that pins are required in a RAM chip for- the Address lines, the Data lines, one Chip Select signal, one Read/Write signal, one Ground line and one Power Supply line, what will be the total no. of pins required in the above chip?
- Consider the following PDP-11 code segment and answer any two of the questions below.

```

mov #100, R0
mov #1000, R1
mov #2000, R2
sub R3, R3
Rpt: cmp (R1), (R2)
bge Lst2
add (R1)+, R3
br Bcount
Lst2: add (R2)+, R3
Bcount: dec R2
bnz Rpt

```

- What task does the code segment perform?
- What will be the length of the machine code for the segment?
- Estimate the execution time of the code segment assuming the memory access time to be 100ns.

2

2

2

TEZPUR UNIVERSITY

Autumn Semester Mid Term (Spl) Examination, 2022

CO 214: Computer Architecture and Organization

Full Marks: 20

Time: 1 Hrs.

Figures in the right hand indicate marks for the individual question.

1. Consider a processor that has 32 registers and 2 addressing modes. The instructions are of 16 bits in length, and these are of 3 types. Type-1 instructions specify 2 operands, Type-2 specifies 1 operand, and the Type-3 are without operand. An operand is specified using a reregister and an addressing mode. There are 14 Type-1 instructions and 64 Type-2 instructions. What is the maximum possible number of instructions of Type-3?

3

2. A 4 Gbit DRAM chip is organized with 4 bits per location and equal number bits for row and column addresses. Determine the size of the refresh buffer.

2

3. A computing hardware is to be designed with a processor generating 32bit address for memory. The memory map in the system is as follows:

EEPROM:	0 to (4M-1)
EPROM:	4M to (512M-1)
RAM:	512M to (4G-1)

$$\begin{array}{r} 64 \\ - 14 \\ \hline 50 \end{array} \quad \begin{array}{r} 64 \\ + 50 \\ \hline 114 \end{array}$$

What will be the Boolean expressions for the signals *EEPROM_Select*, *EPROM_Select*, and *RAM_Select* in the memory unit of the system.

3

4. A computer system with 32 bits address has a cache of 2MB. The cache lines are of 32 bytes. Determine the comparator size for the cache for the following mappings:

- a. Associative mapping - 8 bits
b. 8-way set-associative mapping.

2

$$\begin{array}{r} 64 \\ - 1 \\ \hline \end{array}$$

5. The cache lines in the cache in a computer system can hold 16-byte memory blocks. The cache access time is 5ns. The transfers from the main memory are done in words of 4 bytes. The transfer of the first word of a block from memory takes 70ns and the subsequent words of the block take 10ns each. The average access time of a page from online storage is 2ms. A computer program experiences hit ratios of 0.9 in the cache and 0.999 in the main memory. Determine the average memory access time during the execution. 4
6. A CPU generates 32-bit virtual addresses. The physical memory is of 1GB. The page size used is 4 KB. The MMU has a translation look-aside buffer (TLB) is 4-way set associative and can hold a total of 256 page-table entries. The TLB entries hold the tag for the page number, the frame number, and the 6 control bits. What will be size of the TLB. 3
7. A computer system uses 2-level page tables for virtual to physical address translation. Modules of both the levels of the page table are stored as pages in the main memory. The width of the virtual and physical addresses are 32 bits and 30 bits respectively. The memory is byte addressable, and the pages are of 4KB. The page table entries in both the levels of page tables are 4 bytes wide. Determine the Lengths of the pointers for the level 1 and level 2 of the page table. 2